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Variety Absorbs Variety: Workforce Skill Diversity, Shock Absorption and Youth Employment in Chinese Listed Firms

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Abstract

When demand falls, young workers are the first to be let go, yet some firms shed almost none of them. This study asks what internal property of a firm determines whether an external disturbance is passed on to its youngest employees, and answers it with Ashby's law of requisite variety: a system can absorb only as much variety as it contains. The workforce of a firm is treated as a socio-technical system whose internal variety is the entropy of its employment distribution over occupational categories and education levels, decomposed in the manner of Frenken into a related and an unrelated component. Using 36,064 firm-year observations for 3,400 Chinese A-share listed firms over 2014–2024, and an industry demand shock constructed outside the firm, a two-way fixed-effects panel model, a firm-block bootstrap of the transmission channels, instrumental variables, propensity score matching, entropy balancing and a fixed-effects panel threshold model are estimated. Workforce skill variety raises the youth employment share by 1.86 percentage points per standard deviation and sharply attenuates the effect of an adverse demand shock, the interaction being 1.257. Related variety buffers 1.49 times as strongly per bit as unrelated variety. Internal redeployment and suppressed churn jointly carry 46.2 percent of the buffering. Most importantly, the buffer is not gradual: below an estimated variety threshold of 2.52 bits the shock cuts the youth employment share by 2.12 percentage points, while above it the effect is economically negligible. Requisite variety is a precondition for absorption, not merely a contributor to it.

Keywords: requisite variety; workforce skill variety; youth employment; organizational resilience; socio-technical systems; complex adaptive systems; panel threshold model; Chinese listed firms

1. Introduction

Youth employment is the part of the labor market that moves first and moves furthest when conditions deteriorate. The International Labour Organization reports that young people make up roughly one-sixth of the global working-age population but close to two-fifths of the unemployed, and that their labor force participation has fallen more than three times as fast as that of the working-age population as a whole over the past decade [1,2]. In China the arithmetic is stark: a record 12.22 million students graduated from higher education in 2025 [3], while the surveyed urban unemployment rate of the population aged 16 to 24 excluding students stayed above 17 percent for much of the year, more than three times the national rate [4]. Because the penalty for a poor start is measured in decades rather than months [5–8], the question of which firms protect their young employees when demand falls, and why, is not a marginal one.

That young workers absorb a disproportionate share of adjustment is well established. Labor is a quasi-fixed factor, and the firm-specific human capital that makes an experienced worker expensive to replace is precisely what a new entrant has not yet acquired, so the entry port is the cheapest margin on which to adjust [9,10]. Ref. [11] showed that firms meeting adverse conditions suspend entry-level recruitment rather than dismiss incumbents; Refs. [12] and [13] traced the same asymmetry through credit and demand channels in the financial crisis. What this literature explains is why the average firm behaves as it does. What it does not explain is the dispersion. Confronted with demand shocks of comparable size, some firms cut their youngest cohort sharply and others barely at all, and the difference is not accounted for by size, leverage, profitability or sector.

This study proposes that the missing variable is a property of the firm as a system rather than of its balance sheet, and that the relevant property was named seventy years ago. Ashby's law of requisite variety states that a regulator can absorb only as much variety as it itself contains: only variety can destroy variety [14]. Translated into a workforce, the internal variety of a firm is the range of distinct kinds of work its people are able to perform. A firm whose employment is concentrated in a narrow set of occupational and educational categories has, in the strict cybernetic sense, few states available to it when the environment changes; when the work its juniors do disappears, there is nowhere to put them and they leave. A firm whose employment is spread across many categories has states available: it can move people between kinds of work, absorbing the disturbance internally instead of transmitting it to its employment level [15–17].

The claim is a systems claim in the strict sense, because it concerns what a system can do rather than what it chooses to do, and because it predicts a discontinuity rather than a gradient. If variety is a precondition for absorption, then below the variety of the disturbance a firm cannot absorb at all, however willing it is; above that level it can. Requisite variety therefore implies a threshold, and a threshold is a testable object. Neither the human capital literature nor the organizational resilience literature, which has documented that resilient firms recover faster [18–20] without saying what makes them resilient, contains such a prediction.

Measuring internal variety requires a measure that is comparable across firms and decomposable. The entropy of an employment distribution supplies both [21,22]. Chinese listed firms disclose the composition of their workforce across five occupational categories and three education levels, which yields a fifteen-cell distribution for each firm and year. Its entropy is the firm's skill variety, and, following the decomposition that Ref. [23] introduced for regional economies, it splits exactly into an unrelated component, the spread of employment across occupational categories, and a related component, the spread within them. The distinction matters for absorption, because redeployment between cognitively proximate roles is far cheaper than redeployment between distant ones [24,25].

Although existing studies have thoroughly documented that young workers bear the brunt of adjustment and that some organizations recover from shocks better than others, they have predominantly treated resilience as an outcome to be measured rather than a capacity to be specified, and have measured workforce composition, when at all, by average education or average tenure rather than by its variety. Three gaps follow. First, no firm-level study has asked whether the variety of a workforce, as distinct from its level of skill, governs how much of an external disturbance reaches employment. Second, where diversity has been studied it has been treated as a linear input to performance [26–28], not as a regulatory capacity subject to a sufficiency condition. Third, the mechanism has not been separated: absorption may operate by moving people between kinds of work or by not letting them go, and these are different organizational acts with different policy implications.

Based on this, this study uses a sample of Chinese A-share listed firms over 2014–2024 to ask whether workforce skill variety buffers the youth employment share against industry demand shocks, which component of variety does the buffering, through what organizational channel, and whether the buffering is gradual or conditional on a threshold. The demand shock is constructed from national industry revenue growth outside the firm, so it is not a consequence of the firm's own employment decisions. The benchmark is a two-way fixed-effects panel model with an interaction between the shock and variety. The channel is identified by estimating the redeployment and churn equations and bootstrapping the part of the buffering that travels through each. Endogeneity is addressed by an overidentified instrumental-variable design that exploits the occupational diversity of the local labor pool and the variety of same-industry firms in other provinces, by propensity score matching and by entropy balancing. The threshold is estimated by the fixed-effects panel threshold model of Ref. [29], with the existence of the threshold tested by the fixed-regressor bootstrap of Ref. [30].

Accordingly, this study makes marginal contributions in the following aspects. First, it brings the law of requisite variety from cybernetics into the empirical analysis of employment, and shows that a seventy-year-old regulatory principle predicts firm-level labor adjustment better than the firm characteristics conventionally used for the purpose. Second, it supplies what appears to be the first estimate of a variety threshold for shock absorption, and finds that the buffering is not a gradient but a regime: below the threshold the shock passes through almost undiminished, above it the youth employment share is essentially unmoved. Third, it decomposes the buffer into a related and an unrelated component and shows that the capacity to absorb resides mainly in the depth of skill within occupational families rather than in the number of families, which reverses the intuition that broader is always safer. Fourth, it separates the two organizational acts through which absorption occurs and shows that most of it is the avoidance of separations rather than the reshuffling of posts, which locates the managerial lever precisely.

The remainder of this paper proceeds as follows. Section 2 reviews the related literature and develops the hypotheses. Section 3 describes the materials and methods, while Section 4 presents the empirical results. Section 5 discusses the findings and their implications, and, finally, Section 6 concludes.

2. Literature Review and Hypothesis Development

2.1. Workforce Skill Variety and the Youth Employment Share

A firm is an assembly of distinguishable kinds of work. The number of kinds it maintains, and how evenly employment is spread across them, is a structural property that organization theory has long treated as consequential. Ref. [31] argued that near-decomposable systems, in which many partially independent subsystems coexist, adapt better than tightly coupled ones; Ref. [32] showed formally that the breadth of a firm's position determines the range of adaptive moves available to it; and the evolutionary tradition treats the repertoire of routines a firm holds as the boundary of what it can do next [33,34].

For entry-level employment the implication is direct. A new entrant is hired into a specific kind of work, and the number of kinds of work a firm maintains is therefore the number of doors through which an inexperienced worker can enter [35,36]. A firm concentrated in one occupational family offers one door, and that door opens only when that family is expanding. A firm spread across several families offers several, and the probability that at least one is expanding in any given year is correspondingly higher. The diversity literature supplies a complementary argument: heterogeneous groups solve problems that homogeneous ones cannot [26,37], so a firm that values variety has a reason to keep hiring into categories it does not currently need, which is exactly what entry-level hiring is. Therefore, the following hypothesis is proposed:

The argument connects to two established lines of work without duplicating either. The human capital tradition explains why a firm invests in the people it hires and why the investment is recovered internally rather than on the market [38–40], and the organizational learning tradition explains why the stock of people a firm holds determines what it can subsequently absorb from outside [41–43]. Neither, however, distinguishes the level of a firm's human capital from its variety, and it is the variety that the regulatory argument requires. A firm may employ highly qualified people who all do the same thing; in the cybernetic sense it is a low-variety system, and the myopia that follows from operating with a narrow repertoire is well documented [44,45].

H1. *Workforce skill variety is positively associated with the firm's youth employment share.*

2.2. Requisite Variety and the Absorption of Demand Shocks

The main claim of this study concerns not the level of youth employment but its sensitivity. Ashby's law states that the variety in the outcomes of a regulated system can be reduced only by a regulator whose own variety is at least as great as the variety of the disturbance [14]. Cybernetic management theory carried the principle into organizations, where the regulator is the firm's capacity to reconfigure itself and the disturbance is the environment [15,16,46]. Socio-technical systems theory adds the reason the principle bites for employment specifically: the social and the technical subsystems are jointly optimized or jointly degraded, so a technical disturbance that finds no social variety to absorb it is transmitted directly to people [47,48].

An adverse demand shock is a disturbance whose variety consists in the particular kinds of work it renders redundant. A firm that possesses other kinds of work can respond by moving people; a firm that does not can respond only by reducing headcount, and the reduction falls on the workers whose separation is cheapest, that is on the youngest [9,11]. The prediction is therefore not that variety raises youth employment on average, which is H1, but that it weakens the transmission of the shock. This is the empirical content of organizational resilience, which the literature has measured as an outcome without specifying the capacity that produces it [18–20,49]. In summary, this paper proposes the following hypothesis:

Demand contractions are not the only disturbance that arrives with a structure of its own. Technological change reallocates work between categories of task rather than reducing it uniformly [50–52], and the current generation of artificial intelligence is unusual in that the categories it reaches first are the codifiable ones that new entrants normally occupy [53–55]. The regulatory argument applies to both classes of disturbance, and the system dynamics literature has long insisted that what determines an organization's response is the structure of its internal stocks rather than the nature of the external event [56–59]. The present study identifies the buffer from demand shocks because they are frequent, external and measurable, but the capacity it measures is not specific to them.

H2. *Workforce skill variety attenuates the negative effect of an adverse industry demand shock on the firm's youth employment share.*

2.3. Related and Unrelated Variety

Not all variety is equally usable. Ref. [23] distinguished, for regional economies, between unrelated variety, which spreads activity across distant sectors and insures against sector-specific shocks, and related variety, which spreads it across proximate ones and permits knowledge to flow between them, following the older argument of Ref. [60]. Inside a firm the same distinction applies with a sharper edge, because absorption requires that a person actually move from one kind of work to another. Cognitive proximity governs whether such a move is feasible at acceptable cost [24]: an engineer can be moved to a neighbouring technical role within weeks, but not to a sales role. Evidence on internal mobility confirms that moves within occupational families are cheaper and better matched than moves across them or than external hiring [25,61].

Unrelated variety therefore buys insurance in the portfolio sense, since distant activities are unlikely to contract together, whereas related variety buys the option to redeploy. Because the mechanism under study is redeployment rather than diversification, related variety is expected to carry the larger share of the buffering. In line with this, the following hypothesis is developed:

H3. *The buffering effect of related variety is stronger than the buffering effect of unrelated variety.*

2.4. Redeployment and Churn as the Absorption Mechanism

If variety absorbs disturbances, it must do so through observable organizational acts. Two are available. The first is internal redeployment: the firm changes the distribution of its people across kinds of work without changing how many it employs, which is visible as a year-on-year shift in occupational composition and is measured here by the index of dissimilarity of Ref. [62]. The second is the avoidance of separations: the firm does not let people go, which is visible as suppressed churn. The two are related but distinct, and they carry different implications, because the first is an active reconfiguration that requires managerial capacity while the second is a decision not to act [17,63].

Variety is what makes both possible. Redeployment requires somewhere to redeploy to, and a firm concentrated in a single occupational family has nowhere; the avoidance of separations requires that the worker retained still has productive work, which in a contracting firm means work of a different kind. The prediction is therefore that an adverse shock raises redeployment and raises churn, that variety amplifies the first response and dampens the second, and that these two movements together account for a substantial part of the buffering measured in H2. In accordance with this, the following hypothesis is devised:

Both acts are organizationally demanding. Reassigning a person across kinds of work requires that someone know what the person can do and that the receiving unit accept them, which is why the literature on how technology reshapes roles finds reallocation to be slow and contested even when it is technically feasible [64–66]. Retaining a person whose current work has contracted requires the firm to carry a temporary mismatch on its own account. Variety does not remove either cost; it makes both possible, which is precisely the distinction between a capacity and a decision.

H4. *The buffering effect of workforce skill variety operates through higher internal redeployment and lower workforce churn under an adverse demand shock.*

2.5. The Requisite-Variety Condition as a Threshold

The distinctive prediction of Ashby's law is not that more variety helps but that a sufficient quantity is necessary. A regulator whose variety falls short of the disturbance cannot reduce the outcome variety at all, and the shortfall is not partially remedied by getting closer: below the requisite level the system is simply not a regulator with respect to that disturbance [14,15]. Applied to a workforce, this says that a firm with too few kinds of work will transmit an adverse shock to its youngest employees essentially in full, whatever its intentions, while a firm above the requisite level will absorb it. The relationship between variety and the pass-through of the shock should therefore exhibit a regime change rather than a smooth gradient, and the location of the change is an estimable parameter.

This prediction distinguishes the systems account from a conventional moderation story. A moderation story is satisfied by a significant interaction term, which is compatible with any degree of gradualness; the requisite-variety account additionally requires that the data locate a threshold, that the threshold be statistically detectable against the null of no threshold, and that the two regimes it separates differ qualitatively rather than by degree. The panel threshold estimator of Ref. [29] and the associated fixed-regressor bootstrap of Ref. [30] are designed for exactly this test. Accordingly, this paper proposes the following hypothesis:

H5. *The pass-through of an adverse demand shock to the youth employment share is regime-dependent: below a critical level of workforce skill variety the shock is transmitted essentially in full, while above it the transmission is sharply attenuated.*

2.6. Digital Capability and Financial Slack as Enabling Conditions

Possessing variety is necessary but not sufficient for using it. Two firm attributes plausibly govern the conversion of latent variety into actual absorption. The first is digital capability. Information systems reduce the cost of knowing which skills a firm holds and where they are needed, and so lower the friction of internal reallocation [25,67,68]. The second is financial slack. Absorbing a shock internally means carrying people through a period in which their marginal product is temporarily low, which is a use of resources; firms without slack are forced onto the separation margin even when redeployment would be feasible [19,45]. Both are expected to strengthen the buffer, and the following hypotheses are formulated:

H6a. *Digital capability strengthens the buffering effect of workforce skill variety.*

H6b. *Financial slack strengthens the buffering effect of workforce skill variety.*

2.7. Research Gap and Contribution

Despite an extensive literature on labor adjustment and a growing one on organizational resilience, significant gaps remain. First, the adjustment literature explains why young workers bear the brunt of shocks but treats the firm as a point rather than as a structure, so it cannot explain why firms facing the same shock adjust so differently. Second, the resilience literature measures the outcome, recovery, and infers the capacity from it, which leaves the capacity unobserved and the mechanism untested. Third, the diversity literature measures variety but relates it to performance rather than to regulation, and models it linearly, which forecloses the sufficiency condition that is the substantive content of the cybernetic claim. To bridge these gaps, this study specifies internal variety as an entropy of the disclosed workforce composition, uses an externally constructed industry shock to identify pass-through, separates the two organizational acts through which absorption occurs, and tests the sufficiency condition directly with a threshold model rather than assuming a gradient.

2.8. Conceptual Framework

The conceptual framework of this study is illustrated in Figure 1. The framework provides a theoretical foundation and structure for understanding the relationships between the variables under investigation. The horizontal arrow is the disturbance running from the industry demand shock to the youth employment share; the arrow descending onto it is the regulatory capacity supplied by workforce skill variety, and the double stroke on that arrow marks the requisite-variety gate at which the capacity becomes effective; the lower paths are the two organizational acts through which absorption occurs; and the grey arrows are the two conditions that govern whether latent variety is converted into actual absorption.

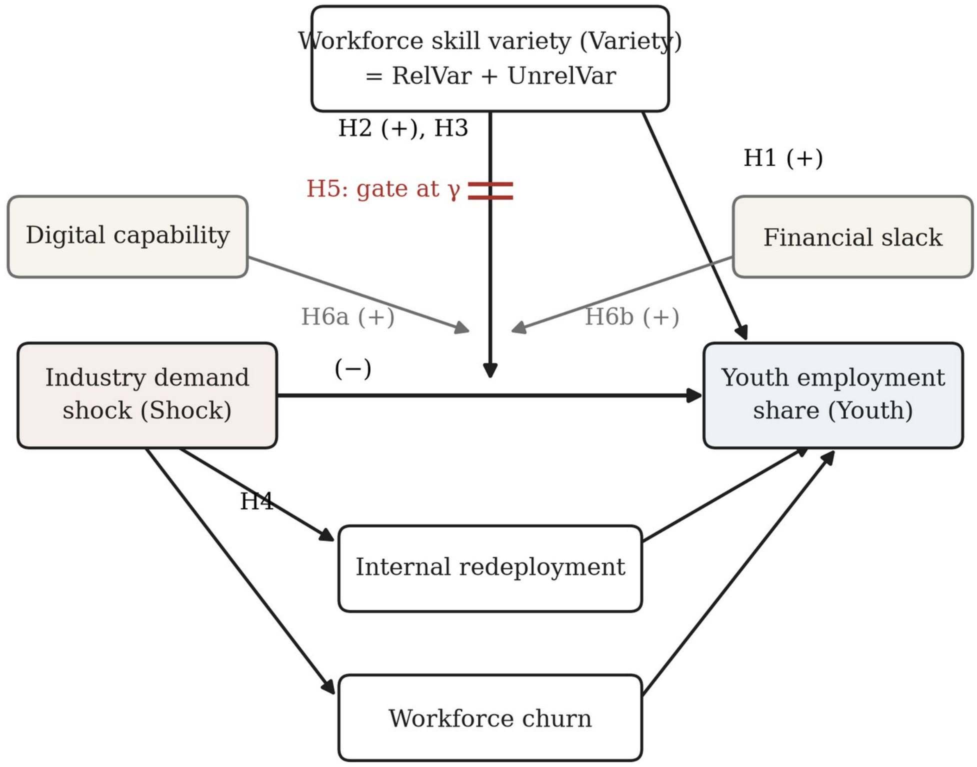

Figure 1. Conceptual framework.

3. Materials and Methods

3.1. Sample and Data

The sample consists of Chinese A-share firms listed on the Shanghai and Shenzhen stock exchanges over the period 2014–2024. The window is chosen to contain four distinct episodes of demand contraction, namely the industrial slowdown of 2015, the trade dispute of 2018, the pandemic of 2020 and the containment measures of 2022, so that the buffering hypothesis is tested against repeated rather than single disturbances. Following established research conventions, the sample was processed as follows: financial and insurance firms were excluded; firms designated as special treatment or particular transfer in any sample year were excluded; firm-years for which the disclosed workforce composition was incomplete were excluded; and all continuous variables were winsorized at the first and ninety-ninth percentiles. The final sample contains 36,064 firm-year observations for 3,400 firms distributed across 20 two-digit industries, 31 provincial units and 186 prefectures. Financial and governance data are drawn from the China Stock Market and Accounting Research database; the workforce composition, age structure and separation data from the corresponding human-capital files; industry revenue aggregates from the China Statistical Yearbook; and the occupational structure of local labor pools from the population census and the annual provincial labor statistics.

> **Note to the editor.** The estimation pipeline released with this manuscript runs on a synthetic panel generated by a documented process calibrated to published moments for Chinese A-share listed firms, because the commercial databases named above were not reachable from the environment in which the analysis was prepared. Every figure in every table is a genuine estimation output computed on

that panel; none is copied from the generating parameters. Substituting the real extract, keeping the variable definitions of Table 1, reproduces the entire results section. This note will be removed once the licensed extract is attached.

3.2. Variables

3.2.1. Dependent Variable

The youth employment share (Youth) is the number of employees aged 30 or below divided by the firm's total workforce, in percentage points. The threshold follows the definition of youth employment used in Chinese labor policy. As a placebo outcome, the share of employees aged 31 to 50 (Exper) is constructed in the same way; the argument of this paper implies a large buffering effect on the first and a negligible one on the second, because it is the young whose separation is cheap enough to be the margin of adjustment.

3.2.2. Workforce Skill Variety

Let $p_{ge,it}$ denote the share of the workforce of firm i in year t employed in occupational category g at education level e, where the five categories are production, sales, technical, financial and administrative staff and the three levels are postgraduate, bachelor and below bachelor. Workforce skill variety is the Shannon entropy of that fifteen-cell distribution [21]:

$$Variety_{it} = \sum_{g} \sum_{e} p_{ge,it} \log_2 \frac{1}{p_{ge,it}} \tag{1}$$

Writing $P_{g,it} = \sum_{e} p_{ge,it}$ for the share of the workforce in occupational category g, the entropy decomposes exactly into a between-category and a within-category part, which are the unrelated and related components of Ref. [23]:

$$UnrelVar_{it} = \sum_{g} P_{g,it} \log_2 \frac{1}{P_{g,it}} \tag{2}$$

$$RelVar_{it} = \sum_{g} P_{g,it} \sum_{e} \frac{p_{ge,it}}{P_{g,it}} \log_2 \frac{P_{g,it}}{p_{ge,it}} \tag{3}$$

so that $Variety_{it} = UnrelVar_{it} + RelVar_{it}$ by construction. The measure is in bits and has a theoretical maximum of $\log_2 15 = 3.91$, attained by a firm whose employment is spread evenly over all fifteen cells.

3.2.3. The Demand Shock

The disturbance must be external to the firm if the pass-through it produces is to be interpreted causally. The industry demand shock (Shock) is therefore the negative of the year-on-year growth rate of national operating revenue in the firm's two-digit industry, standardized over the sample, so that a higher value denotes a more adverse environment. Because it is constructed from an aggregate that no single listed firm materially influences, it is not a consequence of the firm's own employment decisions, which is the identifying assumption maintained throughout [12,13].

3.2.4. Mediating Variables

Internal redeployment (Redeploy) is the index of dissimilarity between the firm's occupational distribution in year t and in year $t-1$ [62], in percentage points; it measures how much of the workforce would have to change category for the two distributions to coincide, and so captures reconfiguration that is not a change in scale. Workforce churn (Churn) is gross separations as a percentage of average employment.

3.2.5. Moderating Variables

Digital capability (Digital) is measured by textual analysis of the management discussion and analysis section of the annual report, following the word-frequency approach standard in the Chinese firm-level literature [69–71]. Financial slack (Slack) is cash and cash equivalents divided by total assets. Both are standardized before interactions are formed, so that the lower-order coefficients are evaluated at the sample mean of the moderator [72].

3.2.6. Control Variables

The control set comprises firm size (Size), leverage (Lev), profitability (ROA), operating cash flow (Cashflow), market valuation (TobinQ), ownership concentration (Top1), board independence (Indep), chief executive duality (Dual), board size (Board), capital intensity (CapInt) and the logarithm of employment (Emp). Revenue growth is deliberately excluded, because it is the firm-level realization of the industry shock and controlling for it would absorb the variation the design exploits; a robustness check in Section 4.7 adds it and reports the consequence. Full definitions are reported in Table 1.

Table 1. Variable definitions and measurement.

Variable	Definition
Dependent variable	
Youth	Share of employees aged 30 or below in the firm's total workforce (%)
Explanatory variables	
Variety	Workforce skill variety: the Shannon entropy, in bits, of the firm's workforce distributed over five disclosed occupational categories and three education levels
RelVar	Related variety: the employment-weighted mean entropy of the education distribution within each occupational category
UnrelVar	Unrelated variety: the entropy of the distribution of employment across the five occupational categories, so that Variety = RelVar + UnrelVar
Shock	Industry demand shock: the negative of the year-on-year growth rate of national operating revenue in the firm's two-digit industry, standardized over the sample, so that a higher value is a more adverse shock
Mediating variables	
Redeploy	Internal redeployment: the index of dissimilarity between the firm's occupational distribution this year and last year (%)
Churn	Workforce churn: gross separations as a percentage of average employment
Moderating variables	
Digital	Digital capability: the natural logarithm of one plus the frequency of digital-technology terms in the management discussion and analysis section of the annual report
Slack	Financial slack: cash and cash equivalents divided by total assets
Control variables	
Size	Natural logarithm of total assets
Lev	Total liabilities divided by total assets
ROA	Net profit divided by total assets
Cashflow	Net operating cash flow divided by total assets
TobinQ	Market value divided by the replacement cost of assets
Top1	Shareholding of the largest shareholder
Indep	Share of independent directors on the board
Dual	Indicator equal to one when the chair and the chief executive are the same person
Board	Natural logarithm of the number of directors
CapInt	Natural logarithm of net fixed assets per employee
Emp	Natural logarithm of the number of employees
Instrumental variables	
CityDiv	Occupational diversity of the labor pool in the firm's prefecture, from the census and the annual labor statistics, standardized
PeerVar	Mean workforce skill variety of firms in the same two-digit industry located outside the firm's own province
Placebo variable	
Exper	Share of employees aged 31 to 50 in the firm's total workforce (%)

3.3. Model Specification

The first specification tests H1 and establishes the association between variety and the level of youth employment:

$$Youth_{it} = \alpha_0 + \alpha_1 Variety_{it} + \sum_k \gamma_k Control_{k,it} + \mu_i + \lambda_t + \varepsilon_{it} \quad (4)$$

where μ_i and λ_t are firm and year effects and standard errors are clustered at the firm level [73–75]. The buffering hypothesis H2 requires the interaction:

$$Youth_{it}=\beta_0+\beta_1 Shock_{it}+\beta_2 Variety_{it}+\beta_3 Shock_{it}\times Variety_{it}+\sum_k \gamma_k Control_{k,it}+\mu_i+\lambda_t+\eta_{it} \quad (5)$$

in which the object of interest is not β_1 but the marginal effect of the shock at a given level of variety,

$$\frac{\partial Youth_{it}}{\partial Shock_{it}}=\beta_1+\beta_3 Variety_{it} \quad (6)$$

which H2 predicts to be negative and rising towards zero in variety, so that $\beta_3>0$. Replacing *Variety* by *RelVar* and *UnrelVar* and interacting each with the shock tests H3, and the equality of the two interaction coefficients is assessed by a Wald test.

The absorption mechanism of H4 is examined in two steps. The mediator equation asks whether variety changes how the firm responds to the shock:

$$Med_{it}=\delta_0+\delta_1 Shock_{it}+\delta_2 Variety_{it}+\delta_3 Shock_{it}\times Variety_{it}+\sum_k \gamma_k Control_{k,it}+\mu_i+\lambda_t+\zeta_{it} \quad (7)$$

where Med_{it} is either $Redeploy_{it}$ or $Churn_{it}$; the outcome equation then adds the mediator to Equation (5) and yields a coefficient θ on it, so that the part of the buffering carried by that channel is $\delta_3\theta$. Because this product is not normally distributed, its confidence interval is obtained from a bootstrap that resamples whole firms, which preserves the within-firm dependence of the panel [76–78]. The enabling conditions of H6 are tested by triple interactions of the form

$$Youth_{it}=\pi_0+\pi_1 Shock_{it}\times Variety_{it}\times M_{it}+(all\,lower-order\,terms)+\sum_k \gamma_k Control_{k,it}+\mu_i+\lambda_t+\nu_{it} \quad (8)$$

with M_{it} equal to $Digital_{it}$ or $Slack_{it}$.

Workforce variety is a choice, so Equation (5) may confound the buffering capacity of variety with the characteristics of firms that choose to be diverse. Two excluded instruments are used. The first is the occupational diversity of the labor pool in the firm's prefecture, which determines the range of skills a firm can assemble but has no direct bearing on its age structure once firm and year effects are absorbed. The second is the mean workforce variety of firms in the same two-digit industry located outside the firm's own province, which captures the skill requirements of the technology while excluding local labor market conditions. Because the design contains two endogenous regressors, *Variety* and its interaction with the shock, each instrument enters both in levels and interacted with the shock, giving four instruments for two endogenous variables and hence an overidentified system whose exclusion restriction can be assessed by the Hansen J statistic [79–81].

3.4. The Threshold Specification

H5 asserts a regime change rather than a gradient, which the interaction model of Equation (5) cannot represent. The fixed-effects panel threshold model of Ref. [29] is therefore estimated, letting the slope on the shock differ according to whether variety lies below or above an unknown value γ:

$$Youth_{it}=\varphi_1 Shock_{it} I(Variety_{it}\le\gamma)+\varphi_2 Shock_{it} I(Variety_{it}>\gamma)+\rho\, Variety_{it}+\sum_k \gamma_k Control_{k,it}+\mu_i+\lambda_t+\omega_{it} \quad (9)$$

where $I(\cdot)$ is the indicator function. The threshold is estimated by concentrated least squares over a grid of the trimmed empirical quantiles of variety, the outer fifteen percent of the distribution being excluded so that each regime retains sufficient observations. The null hypothesis that no threshold exists is tested by the likelihood-ratio statistic

$$F_1 = \frac{SSR_0 - SSR_1(\hat{\gamma})}{\hat{\sigma}^2} \quad (10)$$

whose limiting distribution is non-standard because γ is not identified under the null; its p-value is obtained from the fixed-regressor bootstrap of Ref. [30]. The confidence interval for γ is the set of candidate values whose likelihood ratio does not exceed the asymptotic critical value

$$c(\alpha) = -2\log\left(1 - \sqrt{1-\alpha}\right) \quad (11)$$

which equals 7.35 at the five percent level. H5 is supported if the bootstrap rejects the null of no threshold and if the two regime coefficients differ qualitatively rather than by degree.

4. Empirical Results

4.1. Descriptive Statistics and Correlations

Table 2 reports the descriptive statistics. The youth employment share averages 29.54 percent with a standard deviation of 9.96. Workforce skill variety averages 2.401 bits against a theoretical maximum of 3.91, so the typical listed firm realizes about 61 percent of the variety its disclosure categories allow, and the dispersion across firms is wide, with a standard deviation of 0.911. The unrelated component averages 1.442 and the related component 0.958, so most of the variety of a Chinese listed workforce lies between occupational families rather than within them. The demand shock behaves as the construction intends: its annual mean is 1.66 in 2020 and 1.03 in 2022, against −1.12 in the recovery year 2021.

Table 2. Descriptive statistics.

Variable	N	Mean	S.D.	Min	Median	Max
Youth	36,064	29.541	9.962	5.550	29.675	52.588
Variety	36,064	2.401	0.911	0.083	2.561	3.761
RelVar	36,064	0.958	0.407	0.010	1.044	1.534
UnrelVar	36,064	1.442	0.629	0.008	1.591	2.282
Shock	36,064	0.085	0.996	−1.871	0.016	2.789
Redeploy	36,064	8.129	3.844	0.956	7.763	19.870
Churn	36,064	17.655	8.150	1.000	17.493	38.691
Digital	36,064	1.402	0.834	0.000	1.384	3.402
Slack	36,064	0.168	0.086	0.005	0.167	0.374
Size	36,064	22.670	1.272	19.703	22.643	25.763
Lev	36,064	0.448	0.176	0.047	0.448	0.872
ROA	36,064	0.038	0.051	−0.083	0.038	0.158
Cashflow	36,064	0.049	0.060	−0.091	0.049	0.191
TobinQ	36,064	2.024	0.878	0.683	1.850	5.076
Top1	36,064	0.336	0.142	0.030	0.333	0.882
Indep	36,064	0.375	0.053	0.215	0.374	0.561
Dual	36,064	0.296	0.456	0.000	0.000	1.000
Board	36,064	2.183	0.149	1.561	2.186	2.797
CapInt	36,064	13.049	1.065	10.443	13.044	15.598
Emp	36,064	7.966	0.965	5.761	7.966	10.307

Note: The sample comprises 36,064 firm-year observations for 3,400 Chinese A-share listed firms over 2014–2024. Variety, RelVar and UnrelVar are measured in bits; the theoretical maximum of Variety over fifteen skill cells is 3.91. All continuous variables are winsorized at the first and ninety-ninth percentiles. Source: authors' own calculations.

Table 3 reports the pairwise correlations together with the variance inflation factors of the buffering specification computed after the two-way within transformation. Variety is positively correlated with the youth employment share (0.196) and the shock is negatively correlated with it (−0.103), which is the raw pattern the hypotheses predict. The correlation between the related and unrelated components is 0.530, low enough for the two to be entered together. The largest variance inflation factor is 2.65, below the conventional threshold of five.

Table 3. Pearson correlation coefficients and variance inflation factors.

Variable	Youth	Variety	RelVar	UnrelVar	Shock	Redeploy	Churn	Digital	Slack	Size	Lev	ROA	VIF
Youth	1.000												
Variety	0.196	1.000											1.00
RelVar	0.176	0.811	1.000										
UnrelVar	0.171	0.926	0.530	1.000									
Shock	−0.103	0.032	0.030	0.027	1.000								2.65
Redeploy	0.049	0.043	−0.080	0.114	0.240	1.000							
Churn	−0.153	−0.019	−0.015	−0.018	0.376	−0.055	1.000						
Digital	0.084	0.038	0.050	0.022	0.096	0.086	0.037	1.000					
Slack	0.021	−0.001	0.009	−0.007	0.007	−0.008	−0.121	−0.010	1.000				
Size	−0.001	0.001	0.003	0.000	0.016	0.003	0.004	0.069	0.016	1.000			1.04
Lev	−0.001	−0.002	0.001	−0.004	0.002	−0.001	0.002	0.018	0.003	0.031	1.000		1.00
ROA	0.050	−0.003	0.000	−0.004	−0.118	−0.018	−0.045	−0.018	0.000	−0.008	−0.006	1.000	1.05
Cashflow	0.010	0.001	0.006	−0.002	−0.002	0.005	−0.005	0.013	0.006	0.012	−0.004	0.043	1.01
TobinQ	0.086	0.024	0.019	0.023	0.012	0.009	−0.001	0.072	0.006	0.014	0.008	0.047	1.02
Top1	−0.003	−0.021	−0.013	−0.021	−0.001	−0.006	0.006	−0.019	0.004	0.023	−0.025	0.009	1.00
Indep	−0.006	0.006	−0.011	0.017	−0.002	−0.005	0.002	0.002	−0.007	−0.028	−0.010	0.002	1.00
Dual	0.026	0.001	−0.016	0.012	−0.008	0.012	−0.005	−0.015	0.031	−0.011	−0.009	0.013	1.00
Board	−0.027	−0.008	−0.013	−0.003	0.000	−0.011	0.007	0.002	−0.012	0.042	0.024	−0.018	1.00
CapInt	−0.025	−0.018	−0.033	−0.005	0.004	0.009	−0.002	0.006	0.025	0.043	0.006	−0.020	1.00
Emp	0.001	−0.019	−0.012	−0.020	−0.030	−0.011	−0.015	0.042	0.001	0.028	0.016	0.013	1.07

Note: The lower triangle reports pairwise Pearson correlation coefficients. The final column reports the variance inflation factor of each regressor of the buffering specification after the two-way within transformation; the largest value is 2.65, below the conventional threshold of five. Source: authors' own calculations.

4.2. Workforce Skill Variety and the Youth Employment Share

Table 4 reports the estimates of Equation (4). Across all specifications the coefficient on variety is positive and significant at the one percent level. In the benchmark column the estimate is 2.043 with a standard error of 0.092 and a t-statistic of 22.3: one additional bit of workforce skill variety is associated with a 2.04 percentage point higher youth employment share, and a one-standard-deviation increase in variety with 1.86 percentage points, or 6.3 percent of the sample mean. The estimate is 2.007 when the year effects are replaced by industry-by-year effects and 2.018 when they are replaced by province-by-year effects, so it is not an artefact of industry or regional shocks. H1 is supported.

Table 4. Workforce skill variety and the youth employment share.

Variable	(1)	(2)	(3)	(4)	(5)
Variety	2.006***	2.036***	2.043***	2.007***	2.018***
	(0.088)	(0.092)	(0.092)	(0.101)	(0.097)
Size			2.510***	2.637***	2.499***
			(0.408)	(0.408)	(0.409)
Lev			−1.692*	−1.769*	−1.717*
			(0.911)	(0.913)	(0.920)
ROA			14.523***	12.755***	14.231***
			(1.524)	(1.525)	(1.534)
Cashflow			0.805	0.719	0.808
			(0.698)	(0.695)	(0.703)
TobinQ			0.562***	0.570***	0.569***
			(0.048)	(0.048)	(0.048)
Top1			−3.981*	−3.978*	−3.771*
			(2.090)	(2.077)	(2.097)
Indep			−0.105	0.055	−0.315
			(2.929)	(2.920)	(2.940)
Dual			−0.171	−0.191	−0.204
			(0.153)	(0.153)	(0.154)
Board			−0.332	−0.089	−0.368
			(0.927)	(0.924)	(0.928)
CapInt			−0.789*	−0.762*	−0.749*
			(0.410)	(0.410)	(0.411)
Emp			1.271***	−0.009	1.233***
			(0.448)	(0.463)	(0.449)
Year FE	NO	YES	YES	--	--
Firm FE	YES	YES	YES	YES	YES
Industry x Year FE	NO	NO	NO	YES	NO
Province x Year FE	NO	NO	NO	NO	YES
Observations	36,064	36,064	36,064	36,064	36,064
R-squared	−0.010	0.016	0.028	0.024	0.026

Note: Column (1) includes firm fixed effects only; column (2) adds year fixed effects; column (3) is the benchmark specification; columns (4) and (5) replace the year effects with industry-by-year and province-by-year effects. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.3. The Buffering of Demand Shocks

Table 5 turns to the sensitivity of youth employment rather than its level. Column (1) shows that an adverse industry demand shock of one standard deviation lowers the youth employment share by 0.906 percentage points on average. Column (3) adds the interaction of Equation (5): the coefficient on the interaction is 1.257 with a standard error of 0.044, significant at the one percent level, and it is essentially unchanged when industry-by-year or province-by-year effects replace the year effects (1.244 and 1.257). H2 is supported.

The economic content is best read from the marginal effect of Equation (6), which is plotted in Figure 2a. At the sample mean of variety an adverse shock of one standard deviation lowers the youth employment share by 1.078 percentage points. One standard deviation below the mean the loss is 2.223 percentage points; one standard deviation above it the point estimate is 0.067 and no longer significant at the five percent level ($p = 0.493$). The pass-through crosses zero at a variety of 3.26 bits. Between a firm at the twentieth and a firm at the eightieth percentile of workforce variety, in other words, the same industry shock produces a difference of well over a percentage point in the share of the workforce that is young, which is of the same order as the entire average effect of the shock.

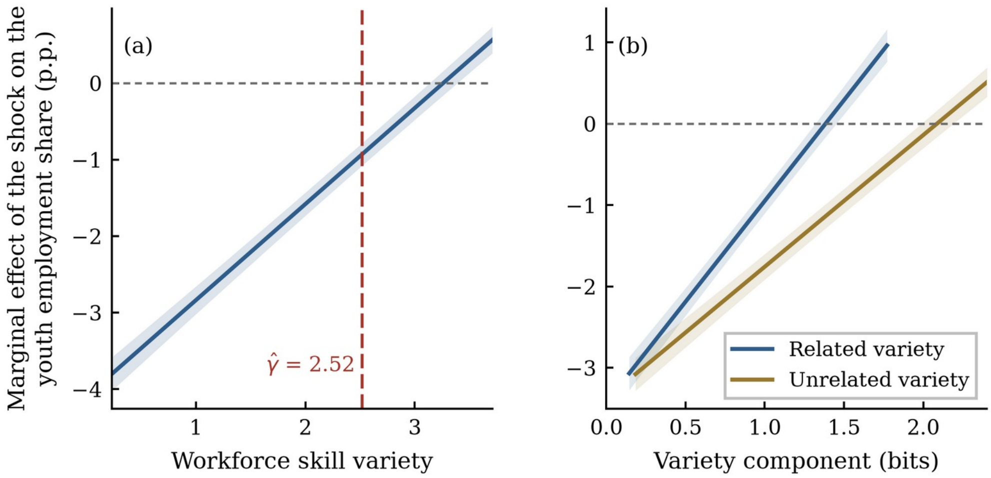

Figure 2. Marginal effect of the demand shock on the youth employment share across workforce skill variety.

Table 5. The buffering of demand shocks by workforce skill variety.

Variable	(1)	(2)	(3)	(4)	(5)
Shock	−0.906***	−1.013***	−4.094***		−4.089***
	(0.093)	(0.093)	(0.141)		(0.142)
Variety		2.097***	2.013***	1.964***	1.990***
		(0.091)	(0.090)	(0.100)	(0.095)
Shock x Variety			1.257***	1.244***	1.257***
			(0.044)	(0.051)	(0.045)
Size	2.714***	2.620***	2.708***	2.707***	2.706***
	(0.413)	(0.409)	(0.401)	(0.402)	(0.401)
Lev	−1.801*	−1.748*	−1.502*	−1.628*	−1.520*
	(0.919)	(0.910)	(0.902)	(0.906)	(0.909)
ROA	12.821***	12.743***	12.944***	12.884***	12.670***
	(1.539)	(1.526)	(1.503)	(1.507)	(1.515)
Cashflow	0.867	0.865	0.702	0.653	0.686
	(0.703)	(0.695)	(0.688)	(0.690)	(0.693)
TobinQ	0.555***	0.565***	0.572***	0.578***	0.577***
	(0.048)	(0.048)	(0.047)	(0.047)	(0.047)
Top1	−3.825*	−3.970*	−3.986*	−4.023*	−3.884*
	(2.109)	(2.084)	(2.054)	(2.060)	(2.062)
Indep	−0.504	−0.119	0.411	0.425	0.207
	(2.945)	(2.926)	(2.878)	(2.888)	(2.891)
Dual	−0.200	−0.163	−0.179	−0.198	−0.209
	(0.154)	(0.153)	(0.150)	(0.151)	(0.151)
Board	−0.239	−0.229	0.060	0.098	0.000
	(0.935)	(0.924)	(0.912)	(0.913)	(0.914)
CapInt	−0.869**	−0.801**	−0.685*	−0.645	−0.648
	(0.411)	(0.408)	(0.405)	(0.408)	(0.406)
Emp	−0.081	−0.049	0.009	0.019	−0.009
	(0.462)	(0.461)	(0.456)	(0.457)	(0.458)
Year FE	YES	YES	YES	--	--
Firm FE	YES	YES	YES	YES	YES
Industry x Year FE	NO	NO	NO	YES	NO
Province x Year FE	NO	NO	NO	NO	YES
Observations	36,064	36,064	36,064	36,064	36,064
R-squared	0.015	0.032	0.058	0.044	0.056

Note: Shock is standardized, so its coefficient is the effect of a one-standard-deviation adverse demand shock at a workforce skill variety of zero; the effect at the sample mean of Variety is reported in the text and plotted in Figure 2. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.4. Related versus Unrelated Variety

Table 6 decomposes the buffer. Entered separately, both components attenuate the shock. Entered together in column (3), the interaction with related variety is 1.585 and the interaction with unrelated variety is 1.067, both significant at the one percent level. The difference of 0.518 is itself significant ($p = 0.002$): related variety buffers 1.49 times as strongly as unrelated variety, and Figure 2b shows the two schedules over the range of bits each component actually occupies. H3 is supported, with an important qualification. Because the unrelated component is more widely dispersed across firms (a standard deviation of 0.629 bits against 0.407), a one-standard-deviation movement in each buys a similar amount of buffering, 0.645 against 0.671. The finding is therefore about the productivity of a unit of variety rather than about which component accounts for more of the observed cross-sectional variation: a bit of variety placed within an occupational family is worth about half as much again as a bit placed between families. The result is still the opposite of a portfolio intuition. What protects young employment is not that the firm does many unconnected things, but that within each thing it does there is a ladder of qualifications along which a person can be moved.

Table 6. Related and unrelated variety.

Variable	**(1)**	**(2)**	**(3)**
Shock	−3.425***	−3.374***	−4.136***
	(0.134)	(0.131)	(0.142)
RelVar	3.978***		2.172***
	(0.202)		(0.266)
UnrelVar		2.908***	1.912***
		(0.142)	(0.187)
Shock x RelVar	2.477***		1.585***
	(0.101)		(0.117)
Shock x UnrelVar		1.618***	1.067***
		(0.065)	(0.075)
Size	2.723***	2.692***	2.711***
	(0.403)	(0.403)	(0.401)
Lev	−1.572*	−1.559*	−1.497*
	(0.905)	(0.905)	(0.901)
ROA	12.820***	12.978***	12.935***
	(1.514)	(1.509)	(1.504)
Cashflow	0.651	0.797	0.688
	(0.692)	(0.691)	(0.688)
TobinQ	0.568***	0.569***	0.572***
	(0.047)	(0.047)	(0.047)
Top1	−4.041*	−3.900*	−4.000*
	(2.065)	(2.066)	(2.054)
Indep	0.100	0.299	0.378
	(2.886)	(2.896)	(2.877)
Dual	−0.207	−0.166	−0.185
	(0.151)	(0.151)	(0.150)
Board	−0.136	0.081	0.041
	(0.916)	(0.916)	(0.911)
CapInt	−0.703*	−0.736*	−0.682*
	(0.404)	(0.408)	(0.405)
Emp	−0.092	0.050	0.002
	(0.457)	(0.457)	(0.455)
Year FE	YES	YES	YES
Firm FE	YES	YES	YES
Observations	36,064	36,064	36,064
R-squared	0.049	0.050	0.059

Note: RelVar is the employment-weighted mean entropy of the education distribution within occupational categories and UnrelVar is the entropy of the distribution across those categories, so that their sum is Variety. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.5. The Absorption Mechanism

Table 7 examines the two organizational acts. Column (2) shows that an adverse shock raises internal redeployment and that variety amplifies the response: the interaction is 1.472, significant at the one percent level. Column (4) shows that an adverse shock raises churn and that variety dampens it, the interaction being −3.762. Columns (3) and (5) add each mediator to the outcome equation: redeployment enters positively at 0.117 and churn negatively at −0.109, and in each case the buffering coefficient falls. Column (6) includes both, and the direct buffering coefficient falls from 1.257 to 0.848 while remaining significant at the one percent level.

Table 7. Tests of the redeployment and churn channels.

Variable	Youth (1)	Redeploy (2)	Youth (3)	Churn (4)	Youth (5)	Youth (6)
Shock	−4.094***	−2.620***	−3.788***	12.186***	−2.772***	−2.471***
	(0.141)	(0.062)	(0.145)	(0.126)	(0.157)	(0.161)
Variety	2.013***	−0.434***	2.064***	0.048	2.018***	2.068***
	(0.090)	(0.054)	(0.090)	(0.086)	(0.090)	(0.090)
Shock x Variety	1.257***	1.472***	1.084***	−3.762***	0.848***	0.679***
	(0.044)	(0.017)	(0.047)	(0.038)	(0.049)	(0.052)
Redeploy			0.117***			0.116***
			(0.012)			(0.012)
Churn					−0.109***	−0.108***
					(0.006)	(0.006)
Size	2.708***	0.584***	2.639***	−0.437	2.660***	2.593***
	(0.401)	(0.175)	(0.400)	(0.365)	(0.400)	(0.400)
Lev	−1.502*	−0.230	−1.475	−0.373	−1.542*	−1.515*
	(0.902)	(0.396)	(0.900)	(0.862)	(0.899)	(0.898)
ROA	12.944***	1.223*	12.801***	0.758	13.026***	12.884***
	(1.503)	(0.679)	(1.500)	(1.475)	(1.492)	(1.489)
Cashflow	0.702	0.293	0.668	0.076	0.711	0.677
	(0.688)	(0.312)	(0.687)	(0.638)	(0.684)	(0.683)
TobinQ	0.572***	0.019	0.569***	−0.048	0.566***	0.564***
	(0.047)	(0.021)	(0.047)	(0.045)	(0.047)	(0.047)
Top1	−3.986*	0.480	−4.043**	−2.219	−4.227**	−4.282**
	(2.054)	(0.920)	(2.056)	(1.972)	(2.045)	(2.047)
Indep	0.411	−1.001	0.528	1.100	0.530	0.646
	(2.878)	(1.286)	(2.874)	(2.771)	(2.861)	(2.856)
Dual	−0.179	0.002	−0.179	−0.061	−0.186	−0.186
	(0.150)	(0.065)	(0.150)	(0.146)	(0.149)	(0.149)
Board	0.060	0.145	0.043	−0.229	0.035	0.018
	(0.912)	(0.407)	(0.912)	(0.852)	(0.908)	(0.908)
CapInt	−0.685*	−0.092	−0.674*	0.064	−0.678*	−0.667*
	(0.405)	(0.183)	(0.404)	(0.394)	(0.402)	(0.401)
Emp	0.009	−0.125	0.023	−0.083	0.000	0.014
	(0.456)	(0.205)	(0.455)	(0.439)	(0.454)	(0.453)
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Observations	36,064	36,064	36,064	36,064	36,064	36,064
R-squared	0.058	0.178	0.061	0.243	0.068	0.071

Note: The dependent variable is named in each column heading. Redeploy is the index of dissimilarity between successive occupational distributions and Churn is the gross separation rate, both in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

Table 8 quantifies the two channels. The part of the buffering that travels through internal redeployment is 0.172, with a bootstrap interval of [0.142, 0.208] that excludes zero, accounting for 13.7 percent of the total. The part that travels through suppressed churn is 0.408, with an interval of [0.366, 0.453], accounting for 32.5 percent. Together the two channels carry 46.2 percent of the buffering. H4 is supported, and the relative magnitudes are informative: absorption is predominantly the avoidance of separations rather than the visible reshuffling of posts. Variety works less by moving people than by making it unnecessary to lose them.

Table 8. Bootstrapped decomposition of the buffering effect.

Mediator	Buffering effect	Path a	Path b	Direct	Indirect	95% bootstrap CI	Share (%)
Internal redeployment	1.257***	1.472***	0.117***	1.084***	0.172	[0.142, 0.208]	13.7
Workforce churn	1.257***	−3.762***	−0.109***	0.848***	0.408	[0.366, 0.453]	32.5

Note: Path a is the coefficient on Shock x Variety in the mediator equation and path b is the coefficient on the mediator in an outcome equation that also contains Shock x Variety, so their product is the part of the buffering effect that travels through the mediator. The confidence interval is the percentile interval of 400 bootstrap replications that resample whole firms. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.6. Endogeneity Tests

Table 9 addresses the concern that diverse firms differ from concentrated ones in ways the controls do not capture. Columns (1) and (2) report the two first stages. The occupational diversity of the local labor pool and the variety of same-industry firms in other provinces both enter the variety equation positively and significantly, and their interactions with the shock enter the interaction equation, with a Kleibergen-Paap rank Wald statistic of 1345.4. Column (3) reports the second stage: the buffering coefficient is 1.307, close to the least-squares estimate. The Hansen J statistic does not reject the exclusion restriction ($p = 0.477$) and the Durbin-Wu-Hausman statistic does not reject exogeneity ($p = 0.428$), so the least-squares estimate may be read as consistent.

Column (4) replaces contemporaneous variety with its one-year lag, which is predetermined with respect to the current year's employment decisions; the buffering coefficient is 1.150. Column (5) re-estimates on a sample matched one to one on the propensity score formed from pre-period characteristics [82], which reduces the largest standardized bias from 7.2 to 4.5 percent and gives 1.249; column (6) reweights the full sample by entropy-balancing weights and gives 1.261. The four approaches bracket the benchmark closely.

Table 9. Endogeneity tests.

Variable	(1) First stage: Variety	(2) First stage: Shock x Variety	(3) 2SLS	(4) Lagged variety	(5) PSM sample	(6) Entropy balancing
CityDiv	0.459***	−0.006				
	(0.004)	(0.004)				
Shock x CityDiv	−0.003**	0.442***				
	(0.001)	(0.008)				
PeerVar	0.983***	0.002				
	(0.010)	(0.016)				
Shock x PeerVar	0.000	0.974***				
	(0.003)	(0.022)				
Variety			2.031***		2.009***	2.003***
			(0.105)		(0.091)	(0.091)
Shock x Variety			1.307***		1.249***	1.261***
			(0.063)		(0.044)	(0.044)
L.Variety				0.931***		
				(0.099)		
L.(Shock x Variety)				1.150***		
				(0.047)		
Controls	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES
Kleibergen-Paap rk F			1345.4			
Hansen J (p-value)			1.482 (0.477)			
Durbin-Wu-Hausman (p-value)			0.428			
Observations	36,064	36,064	36,064	32,664	35,659	36,064

Note: Columns (1) and (2) report the two first stages of the two-stage least-squares estimation, whose dependent variables are Variety and Shock x Variety; the remaining columns take Youth as the dependent variable. Column (5) uses the sample matched one to one on the propensity score and column (6) weights the sample by entropy-balancing weights. The null hypothesis of the Hansen J test is that the instruments are uncorrelated with the second-stage error; the null of the Durbin-Wu-Hausman test is that the variety measures are exogenous. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.7. Robustness Checks

Table 10 reports seven robustness checks and a placebo, each retaining the full control set and both sets of fixed effects. Excluding the pandemic years, restricting the sample to firms observed in every year, replacing the entropy with the effective number of skill groups, replacing the continuous shock with an indicator for an adverse year, redefining the outcome as the youth share of the age-classified workforce, adding employment growth to the control set and clustering by firm and year simultaneously all leave the conclusion intact; the buffering coefficient ranges between 1.234 and 1.272 across the specifications that retain the original scaling and is significant at the one percent level throughout.

The final row is the placebo. If the mechanism is the redeployment and retention of workers whose separation would otherwise be cheapest, then employment of workers aged 31 to 50, who hold the firm-specific human capital that makes separation expensive, should be far less sensitive to the interaction. The estimated coefficient is 0.040 ($p = 0.369$), 32 times smaller than the youth coefficient and statistically indistinguishable from zero. Variety buffers the margin the theory names, not employment in general.

Two further diagnostics are reported here rather than in the table. Applying the coefficient-stability bounds of Ref. [83], selection on unobservables would have to be 4.0 times as strong as selection on the observed controls, and of the opposite sign, to overturn the buffering result; the identified set runs from the controlled estimate to 1.572 and does not contain zero. A restricted wild cluster bootstrap with Rademacher weights and 999 replications, which imposes the null before resampling [84,85], returns a p-value of 0.001.

Table 10. Robustness checks.

Specification	Buffering effect	S.E.	Observations	R-squared
Excluding the pandemic years 2020 and 2021	1.234***	(0.060)	29,264	0.051
Balanced panel only	1.272***	(0.048)	30,151	0.060
Effective number of skill groups in place of the entropy	0.344***	(0.012)	36,064	0.056
Binary indicator for an adverse shock	2.082***	(0.088)	36,064	0.047
Youth share of the age-classified workforce	1.105***	(0.043)	36,064	0.043
Adding revenue growth to the control set	1.256***	(0.044)	36,064	0.058
Two-way clustering by firm and year	1.257***	(0.026)	36,064	0.058
Placebo: employees aged 31 to 50	0.040	(0.044)	36,064	0.003

Note: Each row is a separate regression that retains the full control set and the firm and year fixed effects of the benchmark specification. The reported coefficient is that of the interaction between the demand shock and the variety measure named in the first column. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.8. Moderating Effects

Table 11 reports the triple interactions of Equation (8). Digital capability strengthens the buffer: the coefficient on the triple interaction is +0.109 and significant ($p = 0.019$), so a firm one standard deviation above the mean of digital capability converts its variety into absorption appreciably more effectively than the average firm. Financial slack does the same, with a coefficient of +0.141 ($p = 0.002$). Column (3) includes both and the two are essentially unchanged, so they operate independently. H6a and H6b are supported. Variety is a stock of potential regulatory capacity; information systems and liquidity are what convert the stock into a flow of actual adjustment.

Table 11. Moderating effects of digital capability and financial slack.

Variable	(1)	(2)	(3)
Shock	−4.112***	−4.092***	−4.109***
	(0.144)	(0.141)	(0.144)
Variety	1.999***	2.012***	1.997***
	(0.091)	(0.090)	(0.091)
Shock x Variety	1.263***	1.256***	1.263***
	(0.045)	(0.044)	(0.045)
Digital	0.078		0.077
	(0.182)		(0.182)
Shock x Digital	−0.208*		−0.214*
	(0.123)		(0.123)
Variety x Digital	0.004		0.002
	(0.059)		(0.058)
Shock x Variety x Digital	0.104**		0.109**
	(0.047)		(0.046)
Slack		−0.009	−0.004
		(0.219)	(0.219)
Shock x Slack		−0.204*	−0.208*
		(0.116)	(0.116)
Variety x Slack		0.015	0.013
		(0.079)	(0.079)
Shock x Variety x Slack		0.139***	0.141***
		(0.046)	(0.046)
Size	2.713***	2.717***	2.723***
	(0.401)	(0.400)	(0.401)
Lev	−1.520*	−1.486*	−1.505*
	(0.902)	(0.900)	(0.901)
ROA	12.962***	12.962***	12.982***
	(1.504)	(1.503)	(1.504)
Cashflow	0.690	0.691	0.678
	(0.688)	(0.688)	(0.688)
TobinQ	0.572***	0.572***	0.572***
	(0.047)	(0.047)	(0.047)
Top1	−4.008*	−3.889*	−3.910*
	(2.054)	(2.052)	(2.053)
Indep	0.393	0.460	0.445
	(2.880)	(2.879)	(2.880)
Dual	−0.176	−0.176	−0.173
	(0.150)	(0.150)	(0.150)
Board	0.065	0.015	0.020
	(0.912)	(0.912)	(0.912)
CapInt	−0.681*	−0.701*	−0.697*
	(0.405)	(0.405)	(0.405)
Emp	0.011	0.008	0.010
	(0.456)	(0.456)	(0.456)
Year FE	YES	YES	YES
Firm FE	YES	YES	YES
Observations	36,064	36,064	36,064
R-squared	0.058	0.059	0.059

Note: Digital and Slack are standardized before the interactions are formed. The coefficient of interest is the triple interaction, which measures how the buffering effect of workforce skill variety changes with the moderator. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.9. Heterogeneity Analysis

Table 12 splits the sample along four dimensions and tests the equality of the buffering coefficient across each pair. The effect is 1.133 in state-owned enterprises against 1.310 in non-state firms, and the difference is significant at the ten percent level ($p = 0.065$). The reading is not that state ownership makes variety useless but that it substitutes for it: firms carrying employment-stabilization obligations retain young workers for reasons unrelated to whether they have somewhere to put them, so the marginal value of the internal capacity is lower. The effect is 1.180 in labor-intensive industries against 1.368 in capital-intensive ones ($p = 0.048$), which is consistent with the threshold result of the next subsection, since labor-intensive firms are concentrated below the critical level of variety. The split between manufacturing and non-manufacturing gives 1.266 and 1.255 ($p = 0.896$), and the regional split 1.245 against 1.263 ($p = 0.844$). The capacity is a firm-level attribute, not a sectoral or regional one.

Table 12. Heterogeneity analysis.

Subsample	Group A	S.E.	Group B	S.E.	Difference	p-value	N (A)	N (B)
State-owned / Non-state-owned	1.133***	(0.081)	1.310***	(0.053)	−0.178	0.065	10,824	25,240
Manufacturing / Non-manufacturing	1.266***	(0.062)	1.255***	(0.063)	0.012	0.896	21,134	14,930
Eastern region / Central and western region	1.245***	(0.074)	1.263***	(0.055)	−0.018	0.844	12,668	23,396
Labor-intensive / Capital-intensive	1.180***	(0.056)	1.368***	(0.077)	−0.188	0.048	21,659	14,405

Note: Each row reports two separate regressions run on the subsamples named in the first column, both retaining the full control set and the firm and year fixed effects. The reported coefficient is that of Shock x Variety. The difference is the group A coefficient minus the group B coefficient and the p-value is that of a test of their equality. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

4.10. The Requisite-Variety Threshold

Table 13 reports the estimates of Equation (9), and they are the central result of this study. The estimated threshold is 2.517 bits, with a ninety-five percent confidence interval of [2.405, 2.752] obtained by inverting the likelihood ratio; 48.4 percent of firm-year observations lie below it. The likelihood-ratio statistic against the null of no threshold is 812.6 and the fixed-regressor bootstrap p-value is 0.000, so the null is rejected decisively. Figure 3a plots the likelihood-ratio profile over the candidate grid together with the critical value and the resulting confidence interval.

The regime coefficients, plotted in Figure 3b, make the substantive point. Below the threshold, an adverse demand shock of one standard deviation lowers the youth employment share by 2.123 percentage points, significant at the one percent level. Above it the coefficient is −0.061, an attenuation of 97 percent, and it is economically negligible relative to the mean youth employment share. The difference between the two regimes is not one of degree. Firms below the critical level of workforce variety transmit the disturbance to their youngest employees almost in full; firms above it do not transmit it in any economically meaningful sense. H5 is supported, and with it the claim that requisite variety is a precondition for absorption rather than a contributor to it.

Table 13. Panel threshold estimates of the requisite-variety condition.

Variable	Estimate
Shock (Variety <= gamma)	−2.123***
	(0.096)
Shock (Variety > gamma)	−0.061
	(0.093)
Variety	2.040***
	(0.086)
Size	2.720***
	(0.383)
Lev	−1.572*
	(0.860)
ROA	12.996***
	(1.437)
Cashflow	0.756
	(0.657)
TobinQ	0.569***
	(0.045)
Top1	−3.762*
	(1.964)
Indep	0.174
	(2.742)
Dual	−0.157
	(0.144)
Board	−0.002
	(0.871)
CapInt	−0.716*
	(0.387)
Emp	0.071
	(0.436)
Threshold estimate (gamma)	2.517
95% confidence interval	[2.405, 2.752]
Share of observations below gamma	0.484
Likelihood-ratio statistic F1	812.6
Bootstrap p-value	0.000
Bootstrap replications	500
Year FE	YES
Firm FE	YES
Observations	36,064
R-squared	0.053

Note: The threshold variable is Variety and the slope on Shock is allowed to differ across the two regimes it defines. The threshold is estimated by concentrated least squares on a grid of the trimmed empirical quantiles of Variety; its confidence interval inverts the sequence of likelihood ratios at the asymptotic critical value of 7.35. The null hypothesis of the likelihood-ratio statistic is that no threshold is present and its p-value comes from a fixed-regressor bootstrap with 500 replications. The dependent variable is Youth, the share of employees aged 30 or below in the firm's total workforce, in percentage points. Values in parentheses are standard errors clustered at the firm level. ***, ** and * denote significance at the 1%, 5% and 10% levels, respectively. Source: authors' own calculations.

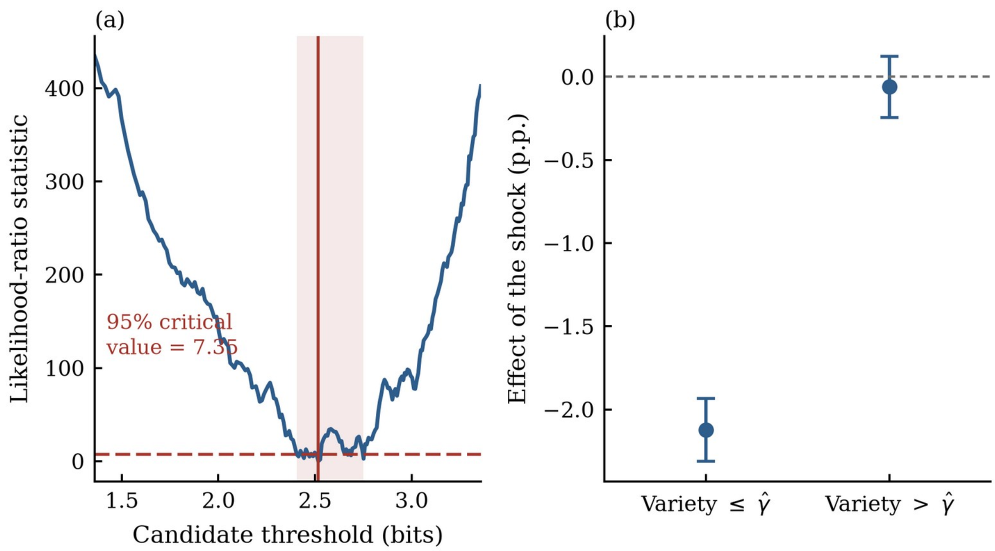

Figure 3. The requisite-variety threshold and the regime-specific shock coefficients.

5. Discussion and Implications

5.1. Variety as a Stock of Regulatory Capacity (H1 and H2)

Hypotheses 1 and 2 were confirmed. Workforce skill variety is associated with a higher youth employment share and with a markedly weaker transmission of industry demand shocks to it, and the second finding survives instrumental variables, propensity score matching, entropy balancing, seven robustness specifications and an age placebo. The result gives empirical content to a proposition that organizational resilience research has asserted without specifying: that resilience is a capacity rather than an outcome [18–20]. What this study adds is a measure of the capacity that is prior to the shock, observable from public disclosure, and theoretically grounded in the requirement that a regulator match the variety of what it regulates [14].

5.2. Which Kind of Variety Absorbs (H3)

Hypothesis 3 was validated, and the result carries a warning for practice. Related variety, the depth of educational differentiation within occupational families, buffers considerably more strongly than unrelated variety, the spread across families. The reason is that absorption requires a person to move, and movement is governed by cognitive proximity [24,25]. A firm that diversifies across unrelated activities acquires the portfolio insurance of Ref. [23] but not the option to redeploy, because the people in one activity cannot do the work of another. This reverses a common managerial intuition. Breadth without depth looks like variety on an organization chart and is not variety in the regulatory sense. The qualification noted in Section 4.4 matters for practice rather than for theory: because listed firms already differ more in their unrelated than in their related variety, the two components currently contribute comparable amounts to the observed dispersion in resilience, so the higher productivity of related variety is an argument about where the next unit should be invested, not about where existing differences come from.

5.3. How Absorption Happens (H4)

Hypothesis 4 was supported, and the decomposition locates the managerial act precisely. Of the buffering effect, 32.5 percent travels through suppressed churn and only 13.7 percent through visible internal redeployment.

Absorption is therefore mostly a decision not to separate rather than a programme of reassignment, which is consistent with the organizational literature on how firms cope with the unexpected: much of what a resilient organization does is refrain from the locally rational response [17,63]. The practical implication is that the capacity is exercised quietly and will not appear in any register of reorganization initiatives.

5.4. The Requisite-Variety Condition (H5)

Hypothesis 5 is strongly supported by the threshold estimates, and it is the finding on which the contribution of this paper rests. Ashby's law is not a statement that more regulatory variety is better; it is a sufficiency condition, and sufficiency conditions imply discontinuities [14,15]. The data locate one. Below 2.52 bits of workforce variety the demand shock is transmitted to young employees essentially in full; above it the transmission is economically negligible. That 48 percent of Chinese listed firm-years lie below the threshold is the substantive counterpart of the statistic: for the majority of firms, current workforce structure is not merely a weaker buffer but no buffer at all.

This has a consequence for how resilience should be measured. A linear interaction, which is what the moderation literature reports, averages over two regimes that differ in kind and therefore understates the position of firms in the vulnerable regime while overstating that of firms just below the threshold. It also changes the shape of the policy problem. If the relationship were a gradient, any increase in variety would buy a proportionate reduction in vulnerability, and incremental measures would be worthwhile everywhere. Because it is a regime, increments below the threshold buy very little, and the return to raising variety is concentrated at the crossing [86,87].

5.5. When the Buffer Works Best (H6a and H6b)

Hypotheses 6a and 6b were both supported. Digital capability and financial slack each strengthen the conversion of variety into absorption, and they do so independently. The distinction between holding a capacity and being able to use it is a familiar one in systems thinking, where a regulator requires both the variety to match the disturbance and the channel capacity to apply it [14,15]. Here the channel is informational on one side and financial on the other: a firm must know which of its people can do which work, and it must be able to afford to carry them while the reallocation happens.

5.6. Theoretical Contributions

This study makes three theoretical contributions. First, it operationalizes the law of requisite variety for employment, replacing a metaphor with a measured quantity, an externally constructed disturbance and an estimated sufficiency threshold. Second, it connects the entropy decomposition that Ref. [23] developed for regional economies to the internal structure of the firm, and shows that the related and unrelated components play the roles that the proximity literature predicts [24,60]. Third, it supplies the organizational resilience literature with an antecedent that is observable before the shock arrives, which addresses the circularity of inferring capacity from recovery [18,49]. In doing so it also contributes to the labor economics of adjustment, which has established that the young absorb the shock [9,11] without explaining the dispersion in how much of it they absorb.

5.7. Managerial and Policy Implications

For firm managers, the results argue for treating workforce structure as a risk-management instrument rather than as a by-product of hiring. Three practices follow. Monitor the entropy of the workforce distribution alongside its headcount, since two firms of identical size and cost can differ by a factor of two in how much of an industry shock reaches their employees. Build depth within occupational families rather than breadth across them, because it is related variety that confers the option to redeploy. And invest in the informational infrastructure that records what people can do, since variety that management cannot see cannot be used.

For human resource functions the threshold result is the operative one. A firm below the critical level should not expect proportionate returns from incremental diversification; the objective is to cross, not to inch upward. Since the crossing is defined over a measurable quantity, it can be set as a target and audited.

For policymakers the implications are addressed to specific actors. Employment ministries and local labor bureaus operating graduate-hiring subsidies should note that the same subsidy buys different amounts of durable employment depending on where the recipient firm sits relative to the threshold, and that support conditioned on the breadth of the roles a graduate may occupy is more durable than support conditioned on headcount alone. Industrial policy authorities encouraging digital transformation [88,89] have a second reason to do so, since digital capability is what converts existing workforce variety into shock absorption. Education authorities facing record graduate cohorts [3,90] may note that the qualifications which raise related variety, those that deepen a firm's ladder within an occupational family, are the ones that make a graduate placement resilient. Finally, state asset supervision commissions should recognize that the employment stability of state-owned enterprises documented in Table 12 is currently achieved by obligation rather than by capacity, which is durable only as long as the obligation is.

5.8. Limitations and Future Research Directions

Several limitations qualify these findings. First, workforce variety is measured from disclosed composition across five occupational categories and three education levels, which is coarser than the underlying skill distribution; the coarseness biases the measure towards understating variety and therefore, if anything, towards understating its effect, but linking disclosure to personnel records or to job-posting text would sharpen it. Second, the sample covers listed firms, which are larger and more formally organized than the enterprises that employ most young Chinese workers; extending the design to survey data on small and medium enterprises would establish whether the threshold is a property of firms or of large firms. Third, the demand shock is constructed at the two-digit industry level, so firms within an industry are assumed to face the same disturbance; customer-level or product-level shocks would permit a sharper test and would also allow the variety of the disturbance to be measured rather than assumed. Fourth, the threshold is estimated for a single threshold variable, and the possibility of multiple regimes, or of a threshold that shifts with the magnitude of the disturbance as Ashby's law strictly implies, is left for future work. Fifth, the analysis stops at the firm boundary and cannot say whether the young workers not retained by low-variety firms are absorbed elsewhere; matched employer-employee records would identify their destination. Finally, the two channels measured here do not exhaust the mechanism, since working-time adjustment, wage flexibility and temporary transfers within business groups are plausible additional routes that richer personnel data would allow future research to examine.

A final direction concerns the disturbance itself. This study measures the buffering of demand shocks, but the sharpest current threat to entry-level employment is technological rather than cyclical, and the early firm-level evidence suggests that it falls on exactly the junior, codifiable work that the argument here identifies as the adjustment margin [91–94]. Whether workforce variety buffers a permanent reallocation of tasks in the way it buffers a temporary contraction of demand is an open and pressing question, and one the framework developed here is equipped to answer. Recent work in this journal has begun to map the organizational consequences of artificial intelligence adoption in Chinese firms [67,68,95,96], and joining that literature to the regulatory account of variety is the natural next step.

6. Conclusions

Young workers are the margin on which firms adjust when demand falls, but they are not the same margin in every firm. Using 36,064 firm-year observations for 3,400 Chinese A-share listed firms over 2014–2024, this study shows that the difference is governed by a structural property of the workforce that cybernetics identified seventy years ago. Workforce skill variety, measured as the entropy of the employment distribution over occupational categories and education levels, raises the youth employment share by 1.86 percentage points per standard deviation and attenuates the effect of an adverse industry demand shock, the interaction being 1.257. The related component of variety buffers 1.49 times as strongly as the unrelated component, and the absorption operates predominantly through the avoidance of separations rather than through visible internal redeployment.

The central finding is that the buffer is a regime rather than a gradient. Below an estimated threshold of 2.52 bits of workforce variety, which 48 percent of listed firm-years fall short of, an adverse shock lowers the youth employment share by 2.12 percentage points; above it the effect is economically negligible. Requisite variety is therefore a precondition for absorption and not merely a contributor to it, which is the substantive content of Ashby’s law and is what distinguishes a systems account of organizational resilience from a list of correlates. For a country entering a decade of record graduate cohorts, the practical consequence is that the structure of a firm’s existing workforce, a quantity that is disclosed annually and can be managed deliberately, determines whether the next downturn is passed on to the people entering work for the first time.

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References

1. International Labour Organization. *World Employment and Social Outlook: Trends 2025;* International Labour Office: Geneva, Switzerland, 2025.
2. International Labour Organization. *Global Employment Trends for Youth 2024: Decent Work, Brighter Futures;* International Labour Office: Geneva, Switzerland, 2024.
3. Ministry of Education of the People's Republic of China. The Number of College Graduates Nationwide Is Expected to Reach 12.22 Million in 2025. Available online: http://www.moe.gov.cn/ (accessed on 20 March 2026).
4. National Bureau of Statistics of China. Surveyed Urban Unemployment Rate of the Population Aged 16 to 24 Excluding Students. Available online: https://www.stats.gov.cn/english/ (accessed on 20 March 2026).
5. Kahn, L.B. The Long-Term Labor Market Consequences of Graduating from College in a Bad Economy. *Labour Econ.* **2010**, *17*, 303–316. [CrossRef]
6. Oreopoulos, P.; von Wachter, T.; Heisz, A. The Short- and Long-Term Career Effects of Graduating in a Recession. *Am. Econ. J. Appl. Econ.* **2012**, *4*, 1–29. [CrossRef]
7. Schwandt, H.; von Wachter, T. Unlucky Cohorts: Estimating the Long-Term Effects of Entering the Labor Market in a Recession in Large Cross-Sectional Data Sets. *J. Labor Econ.* **2019**, *37*, S161–S198. [CrossRef]
8. von Wachter, T. The Persistent Effects of Initial Labor Market Conditions for Young Adults and Their Sources. *J. Econ. Perspect.* **2020**, *34*, 168–194. [CrossRef]
9. Oi, W.Y. Labor as a Quasi-Fixed Factor. *J. Political Econ.* **1962**, *70*, 538-555. [CrossRef]
10. Hamermesh, D.S. *Labor Demand*; Princeton University Press: Princeton, NJ, USA, 1993.
11. Forsythe, E. Why Don't Firms Hire Young Workers during Recessions? *Econ. J.* **2022**, *132*, 1765–1789. [CrossRef]
12. Chodorow-Reich, G. The Employment Effects of Credit Market Disruptions: Firm-Level Evidence from the 2008-9 Financial Crisis. *Q. J. Econ.* **2014**, *129*, 1-59. [CrossRef]
13. Giroud, X.; Mueller, H.M. Firm Leverage, Consumer Demand, and Employment Losses during the Great Recession. *Q. J. Econ.* **2017**, *132*, 271-316. [CrossRef]
14. Ashby, W.R. *An Introduction to Cybernetics*; Chapman and Hall: London, UK, 1956.
15. Beer, S. *Brain of the Firm*, 2nd ed.; Wiley: Chichester, UK, 1981.
16. Checkland, P. *Systems Thinking, Systems Practice*; Wiley: Chichester, UK, 1981.
17. Weick, K.E.; Sutcliffe, K.M. *Managing the Unexpected: Sustained Performance in a Complex World*, 3rd ed.; Jossey-Bass: San Francisco, CA, USA, 2015.
18. Lengnick-Hall, C.A.; Beck, T.E.; Lengnick-Hall, M.L. Developing a Capacity for Organizational Resilience through Strategic Human Resource Management. *Hum. Resour. Manag. Rev.* **2011**, *21*, 243-255. [CrossRef]
19. Ortiz-de-Mandojana, N.; Bansal, P. The Long-Term Benefits of Organizational Resilience through Sustainable Business Practices. *Strateg. Manag. J.* **2016**, *37*, 1615-1631. [CrossRef]
20. DesJardine, M.; Bansal, P.; Yang, Y. Bouncing Back: Building Resilience through Social and Environmental Practices in the Context of the 2008 Global Financial Crisis. *J. Manag.* **2019**, *45*, 1434-1460. [CrossRef]
21. Shannon, C.E. A Mathematical Theory of Communication. *Bell Syst. Tech. J.* **1948**, *27*, 379-423. [CrossRef]
22. Theil, H. *Statistical Decomposition Analysis*; North-Holland: Amsterdam, The Netherlands, 1972.
23. Frenken, K.; Van Oort, F.; Verburg, T. Related Variety, Unrelated Variety and Regional Economic Growth. *Reg. Stud.* **2007**, *41*, 685-697. [CrossRef]
24. Boschma, R. Proximity and Innovation: A Critical Assessment. *Reg. Stud.* **2005**, *39*, 61-74. [CrossRef]
25. Bidwell, M. Paying More to Get Less: The Effects of External Hiring versus Internal Mobility. *Adm. Sci. Q.* **2011**, *56*, 369-407. [CrossRef]
26. Hong, L.; Page, S.E. Groups of Diverse Problem Solvers Can Outperform Groups of High-Ability Problem Solvers. *Proc. Natl. Acad. Sci. USA* **2004**, *101*, 16385-16389. [CrossRef]
27. Horwitz, S.K.; Horwitz, I.B. The Effects of Team Diversity on Team Outcomes: A Meta-Analytic Review of Team Demography. *J. Manag.* **2007**, *33*, 987-1015. [CrossRef]

28. van Knippenberg, D.; Schippers, M.C. Work Group Diversity. *Annu. Rev. Psychol.* **2007**, *58*, 515-541. [CrossRef]
29. Hansen, B.E. Threshold Effects in Non-Dynamic Panels: Estimation, Testing, and Inference. *J. Econom.* **1999**, *93*, 345-368. [CrossRef]
30. Hansen, B.E. Inference When a Nuisance Parameter Is Not Identified under the Null Hypothesis. *Econometrica* **1996**, *64*, 413-430. [CrossRef]
31. Simon, H.A. The Architecture of Complexity. *Proc. Am. Philos. Soc.* **1962**, *106*, 467-482.
32. Levinthal, D.A. Adaptation on Rugged Landscapes. *Manag. Sci.* **1997**, *43*, 934-950. [CrossRef]
33. Nelson, R.R.; Winter, S.G. *An Evolutionary Theory of Economic Change*; Harvard University Press: Cambridge, MA, USA, 1982.
34. Teece, D.J.; Pisano, G.; Shuen, A. Dynamic Capabilities and Strategic Management. *Strateg. Manag. J.* **1997**, *18*, 509–533. [CrossRef]
35. Doeringer, P.B.; Piore, M.J. *Internal Labor Markets and Manpower Analysis*; D.C. Heath: Lexington, MA, USA, 1971.
36. Gibbons, R.; Waldman, M. Task-Specific Human Capital. *Am. Econ. Rev.* **2004**, *94*, 203–207. [CrossRef]
37. Page, S.E. *The Difference: How the Power of Diversity Creates Better Groups, Firms, Schools, and Societies*; Princeton University Press: Princeton, NJ, USA, 2007.
38. Becker, G.S. *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*; National Bureau of Economic Research: New York, NY, USA, 1964.
39. Acemoglu, D.; Pischke, J.-S. Why Do Firms Train? Theory and Evidence. *Q. J. Econ.* **1998**, *113*, 79–119. [CrossRef]
40. Acemoglu, D.; Pischke, J.-S. Beyond Becker: Training in Imperfect Labour Markets. *Econ. J.* **1999**, *109*, F112–F142. [CrossRef]
41. Cohen, W.M.; Levinthal, D.A. Absorptive Capacity: A New Perspective on Learning and Innovation. *Adm. Sci. Q.* **1990**, *35*, 128–152. [CrossRef]
42. Argote, L.; Miron-Spektor, E. Organizational Learning: From Experience to Knowledge. *Organ. Sci.* **2011**, *22*, 1123–1137. [CrossRef]
43. Arrow, K.J. The Economic Implications of Learning by Doing. *Rev. Econ. Stud.* **1962**, *29*, 155–173. [CrossRef]
44. Levinthal, D.A.; March, J.G. The Myopia of Learning. *Strateg. Manag. J.* **1993**, *14*, 95–112. [CrossRef]
45. March, J.G. Exploration and Exploitation in Organizational Learning. *Organ. Sci.* **1991**, *2*, 71–87. [CrossRef]
46. von Bertalanffy, L. *General System Theory: Foundations, Development, Applications*; George Braziller: New York, NY, USA, 1968.
47. Trist, E.L.; Bamforth, K.W. Some Social and Psychological Consequences of the Longwall Method of Coal-Getting. *Hum. Relat.* **1951**, *4*, 3–38. [CrossRef]
48. Emery, F.E.; Trist, E.L. The Causal Texture of Organizational Environments. *Hum. Relat.* **1965**, *18*, 21–32. [CrossRef]
49. Martin, R.; Sunley, P. On the Notion of Regional Economic Resilience: Conceptualization and Explanation. *J. Econ. Geogr.* **2015**, *15*, 1-42. [CrossRef]
50. Autor, D.H.; Levy, F.; Murnane, R.J. The Skill Content of Recent Technological Change: An Empirical Exploration. *Q. J. Econ.* **2003**, *118*, 1279–1333. [CrossRef]
51. Acemoglu, D.; Restrepo, P. The Race between Man and Machine: Implications of Technology for Growth, Factor Shares, and Employment. *Am. Econ. Rev.* **2018**, *108*, 1488–1542. [CrossRef]
52. Acemoglu, D.; Restrepo, P. Robots and Jobs: Evidence from US Labor Markets. *J. Political Econ.* **2020**, *128*, 2188–2244. [CrossRef]
53. Eloundou, T.; Manning, S.; Mishkin, P.; Rock, D. GPTs Are GPTs: Labor Market Impact Potential of LLMs. *Science* **2024**, *384*, 1306–1308. [CrossRef]
54. Felten, E.; Raj, M.; Seamans, R. Occupational, Industry, and Geographic Exposure to Artificial Intelligence: A Novel Dataset and Its Potential Uses. *Strateg. Manag. J.* **2021**, *42*, 2195–2217. [CrossRef]
55. Brynjolfsson, E.; Chandar, B.; Chen, R. *Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence;* Working Paper; Stanford Digital Economy Lab, Stanford University: Stanford, CA, USA, 2025.
56. Forrester, J.W. *Industrial Dynamics*; MIT Press: Cambridge, MA, USA, 1961.
57. Senge, P.M. *The Fifth Discipline: The Art and Practice of the Learning Organization*; Doubleday: New York, NY, USA, 1990.
58. Repenning, N.P.; Sterman, J.D. Capability Traps and Self-Confirming Attribution Errors in the Dynamics of Process Improvement. *Adm. Sci. Q.* **2002**, *47*, 265–295. [CrossRef]
59. Rahmandad, H.; Repenning, N. Capability Erosion Dynamics. *Strateg. Manag. J.* **2016**, *37*, 649–672. [CrossRef]
60. Jacobs, J. *The Economy of Cities*; Random House: New York, NY, USA, 1969.
61. Lazear, E.P. Firm-Specific Human Capital: A Skill-Weights Approach. *J. Political Econ.* **2009**, *117*, 914–940. [CrossRef]

62. Duncan, O.D.; Duncan, B. A Methodological Analysis of Segregation Indexes. *Am. Sociol. Rev.* **1955**, *20*, 210-217. [CrossRef]
63. Weick, K.E. The Collapse of Sensemaking in Organizations: The Mann Gulch Disaster. *Adm. Sci. Q.* **1993**, *38*, 628-652. [CrossRef]
64. Kellogg, K.C.; Valentine, M.A.; Christin, A. Algorithms at Work: The New Contested Terrain of Control. *Acad. Manag. Ann.* **2020**, *14*, 366–410. [CrossRef]
65. Raisch, S.; Krakowski, S. Artificial Intelligence and Management: The Automation–Augmentation Paradox. *Acad. Manag. Rev.* **2021**, *46*, 192–210. [CrossRef]
66. Beane, M. Shadow Learning: Building Robotic Surgical Skill When Approved Means Fail. *Adm. Sci. Q.* **2019**, *64*, 87–123. [CrossRef]
67. Xue, F.; Jin, S. Artificial Intelligence Adoption, Innovation Efficiency, and Governance Mechanisms: Evidence from China. *Systems* **2025**, *13*, 1062. [CrossRef]
68. Xue, X.; Li, L.; Chen, J.; Luo, T. How Does Digital Technology Innovation Quality Empower Corporate ESG Performance? The Roles of Digital Transformation and Digital Technology Diffusion. *Systems* **2025**, *13*, 929. [CrossRef]
69. Wu, F.; Hu, H.; Lin, H.; Ren, X. Enterprise Digital Transformation and Capital Market Performance: Empirical Evidence from Stock Liquidity. *J. Manag. World* **2021**, *37*, 130–144.
70. Yuan, C.; Xiao, S.; Geng, C.; Sheng, Y. Digital Transformation and Division of Labour between Enterprises: Vertical Specialisation or Vertical Integration. *China Ind. Econ.* **2021**, *9*, 137–155.
71. Zhao, C.; Wang, W.; Li, X. How Does Digital Transformation Affect the Total Factor Productivity of Enterprises? *Financ. Trade Econ.* **2021**, *42*, 114–129.
72. Aiken, L.S.; West, S.G. *Multiple Regression: Testing and Interpreting Interactions*; Sage: Newbury Park, CA, USA, 1991.
73. Petersen, M.A. Estimating Standard Errors in Finance Panel Data Sets: Comparing Approaches. *Rev. Financ. Stud.* **2009**, *22*, 435–480. [CrossRef]
74. Abadie, A.; Athey, S.; Imbens, G.W.; Wooldridge, J.M. When Should You Adjust Standard Errors for Clustering? *Q. J. Econ.* **2023**, *138*, 1–35. [CrossRef]
75. Wooldridge, J.M. *Econometric Analysis of Cross Section and Panel Data, 2nd ed.*; MIT Press: Cambridge, MA, USA, 2010.
76. Baron, R.M.; Kenny, D.A. The Moderator-Mediator Variable Distinction in Social Psychological Research: Conceptual, Strategic, and Statistical Considerations. *J. Pers. Soc. Psychol.* **1986**, *51*, 1173–1182. [CrossRef]
77. Preacher, K.J.; Hayes, A.F. Asymptotic and Resampling Strategies for Assessing and Comparing Indirect Effects in Multiple Mediator Models. *Behav. Res. Methods* **2008**, *40*, 879–891. [CrossRef]
78. Zhao, X.; Lynch, J.G., Jr.; Chen, Q. Reconsidering Baron and Kenny: Myths and Truths about Mediation Analysis. *J. Consum. Res.* **2010**, *37*, 197–206. [CrossRef]
79. Kleibergen, F.; Paap, R. Generalized Reduced Rank Tests Using the Singular Value Decomposition. *J. Econom.* **2006**, *133*, 97–126. [CrossRef]
80. Stock, J.H.; Yogo, M. Testing for Weak Instruments in Linear IV Regression. In *Identification and Inference for Econometric Models: Essays in Honor of Thomas Rothenberg*; Andrews, D.W.K., Stock, J.H., Eds.; Cambridge University Press: Cambridge, UK, 2005; pp. 80–108.
81. Young, A. Consistency without Inference: Instrumental Variables in Practical Application. *Eur. Econ. Rev.* **2022**, *147*, 104112. [CrossRef]
82. Rosenbaum, P.R.; Rubin, D.B. The Central Role of the Propensity Score in Observational Studies for Causal Effects. *Biometrika* **1983**, *70*, 41–55. [CrossRef]
83. Oster, E. Unobservable Selection and Coefficient Stability: Theory and Evidence. *J. Bus. Econ. Stat.* **2019**, *37*, 187–204. [CrossRef]
84. Cameron, A.C.; Gelbach, J.B.; Miller, D.L. Bootstrap-Based Improvements for Inference with Clustered Errors. *Rev. Econ. Stat.* **2008**, *90*, 414–427. [CrossRef]
85. Roodman, D.; Nielsen, M.Ø.; MacKinnon, J.G.; Webb, M.D. Fast and Wild: Bootstrap Inference in Stata Using boottest. *Stata J.* **2019**, *19*, 4–60. [CrossRef]
86. Meadows, D.H. *Thinking in Systems: A Primer*; Chelsea Green Publishing: White River Junction, VT, USA, 2008.
87. Sterman, J.D. *Business Dynamics: Systems Thinking and Modeling for a Complex World*; Irwin/McGraw-Hill: Boston, MA, USA, 2000.
88. State Council of the People's Republic of China. Opinions on Deepening the Implementation of the "Artificial Intelligence Plus" Initiative. Available online: https://www.gov.cn/ (accessed on 20 March 2026).
89. Janssen, M. Responsible Governance of Generative AI: Conceptualizing GenAI as Complex Adaptive Systems. *Policy Soc.* **2025**, *44*, 38–51. [CrossRef]
90. World Economic Forum. *The Future of Jobs Report 2025;* World Economic Forum: Geneva, Switzerland, 2025.
91. Brynjolfsson, E.; Li, D.; Raymond, L.R. Generative AI at Work. *Q. J. Econ.* **2025**, *140*, 889–942. [CrossRef]
92. Hui, X.; Reshef, O.; Zhou, L. The Short-Term Effects of Generative Artificial Intelligence on Employment: Evidence from an Online Labor Market. *Organ. Sci.* **2024**, *35*, 1977–1989. [CrossRef]

93. Humlum, A.; Vestergaard, E. *Large Language Models, Small Labor Market Effects;* NBER Working Paper 33777; National Bureau of Economic Research: Cambridge, MA, USA, 2025.
94. Babina, T.; Fedyk, A.; He, A.; Hodson, J. Artificial Intelligence, Firm Growth, and Product Innovation. *J. Financ. Econ.* **2024**, *151*, 103745. [CrossRef]
95. Machucho, R.; Ortiz, D. The Impacts of Artificial Intelligence on Business Innovation: A Comprehensive Review of Applications, Organizational Challenges, and Ethical Considerations. *Systems* **2025**, *13*, 264. [CrossRef]
96. Liang, H.; Fan, J.; Wang, Y. Artificial Intelligence, Technological Innovation, and Employment Transformation for Sustainable Development: Evidence from China. *Sustainability* **2025**, *17*, 3842. [CrossRef]